

JOSEPH GILL

Continuous Improvement Engineer | Manufacturing Automation | AI & Digital Solutions

Jamestown, NY

(814) 558-2767 | Jgill@vertexopsis.net | linkedin.com/in/joseph-gill-4275633a9 | xerozor1-ui.github.io/joseph-gill-portfolio | Vertexopsis.net

PROFESSIONAL SUMMARY

Continuous Improvement Engineer with **6+ years of manufacturing experience** spanning production leadership, quality, fabrication, process improvement, industrial automation, and software development. Proven ability to identify operational bottlenecks and independently develop practical digital and physical solutions to improve productivity, visibility, quality, and decision-making.

Currently designing and deploying **custom manufacturing software, AI applications, ERP integrations, mobile systems, automation tools, analytics platforms, and shop-floor technology** within a full-scale sheet metal fabrication environment. Experienced with Six Sigma methodologies, Kaizen events, KPI development, Epicor ERP, Python development, REST APIs, SQL databases, AI/LLM deployment, industrial hardware, and cross-functional project leadership.

Known for combining direct shop-floor manufacturing knowledge with engineering and software development to build solutions that operators, supervisors, quality, engineering, purchasing, estimating, maintenance, HR, and management can use in real-world production environments.

CORE COMPETENCIES

Continuous Improvement • Kaizen Events • Process Optimization • KPI Development & Reporting • Manufacturing Engineering • Industrial Automation • AI / LLM Applications • Python Software Development • REST API Integration • Epicor ERP Integration • SQL / SQLite Databases • Digital Manufacturing • MES Development • Digital Twin Concepts • Manufacturing Analytics • Workflow Automation • Industrial IoT / IIoT • Raspberry Pi / ESP32 • Android Industrial Devices • Barcode & Material Tracking • OPC UA • Robotic Systems • Quality Systems • Time Studies • Root Cause Analysis • Project Leadership • Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Blackstone Advanced Technologies — Jamestown, NY

Continuous Improvement Engineer

2026 – Present

Lead and support continuous improvement, manufacturing technology, automation, software, and AI initiatives throughout a large custom sheet metal fabrication operation.

- Develop custom software and automation solutions for production, quality, maintenance, engineering, purchasing, estimating, HR, and management.
- Design systems that integrate directly with **Epicor ERP through REST APIs**, allowing real-time manufacturing information to be incorporated into custom applications.
- Perform process analysis, time studies, root cause investigations, and workflow evaluations to identify opportunities for productivity and efficiency improvements.
- Develop KPI reporting and operational dashboards to improve visibility into manufacturing performance.
- Apply **Lean, Six Sigma, and Kaizen methodologies** to manufacturing and administrative processes.
- Lead projects from concept and business justification through software development, testing, operator feedback, implementation, and continuous improvement.
- Work directly with operators, supervisors, engineering, quality, IT, maintenance, and executive leadership to identify problems and develop practical solutions.
- Develop AI-assisted applications while evaluating security, confidentiality, local-model deployment, infrastructure requirements, and organizational AI governance.

- Create internal tools that reduce repetitive administrative work and improve access to manufacturing information.
- Evaluate new manufacturing technologies including industrial scanning, machine monitoring, robotics, autonomous mobile robots, computer vision, AI, IoT sensors, and automated inspection systems.

Selected Engineering & Digital Transformation Projects

Vertex Ops — Custom Manufacturing Operations Platform

- Designed and developed a Python-based manufacturing operations platform integrated with Epicor.
- Created digital workstation and production-zone tracking.
- Developed material staging, kitting, job tracking, quality, maintenance, engineering change, labor, and production-support functionality.
- Designed barcode-driven mobile workflows for operators, material handlers, quality personnel, welders, and maintenance staff.
- Developed concepts for plant-wide digital inventory visibility allowing personnel to locate material, jobs, kits, and production assets quickly.
- Created applications for employee call-ins, vacation workflows, maintenance requests, quality records, and production tracking.

Olympus — AI Orchestration Platform

- Developed an internal AI orchestration environment designed to coordinate specialized manufacturing AI applications.
- Designed distributed workload concepts using local compute resources, GPU systems, Redis queues, worker scheduling, and automated task execution.
- Focused architecture on secure internal AI deployment and reducing reliance on external cloud processing for sensitive manufacturing information.

Atlas — AI-Assisted Estimating & Quoting

- Developed concepts and prototypes for an AI-assisted manufacturing quoting system.
- Integrated engineering drawings, historical quote information, customer files, ERP data, pricing formulas, and quote databases.
- Evaluated vision-language models for interpreting engineering drawings and extracting manufacturing information.

Plutus — Purchasing Analytics

- Developed automated purchasing analytics using live ERP purchasing information.
- Built tools for purchasing analysis, supplier information, raw-material monitoring, and procurement decision support.

Apollo — Equipment Support AI

- Developed an AI knowledge system using equipment manuals and technical documentation to assist personnel with machine troubleshooting and equipment information.

Hephaestus — Welding Knowledge AI

- Developed an AI-assisted welding knowledge platform using internal welding documentation and technical welding resources.

Chronos — Time Study Analytics

- Developed digital time-study and manufacturing-analysis tools to improve routing accuracy and identify opportunities for process improvements.

Soteria — Safety & Improvement Platform

- Developed a centralized application for safety observations, near misses, unsafe conditions, continuous improvement opportunities, and safety-related reporting.

Autonomous Material Movement

- Developed workflow concepts for integrating autonomous mobile robots into material handling and production flow.
- Designed station concepts using sensors, custom fabricated structures, embedded computing, and manufacturing software integration.

Industrial Mobile Scanning

- Designed Android-based barcode workflows for material handlers, quality personnel, welders, and maintenance technicians.
- Developed concepts for asset scanning, material tracking, kit location, maintenance history, and digital inventory management.

Machine Data / IIoT

- Investigated OPC UA, Raspberry Pi, sensor systems, and machine interfaces for capturing production and condition-monitoring information from manufacturing equipment.

Blackstone Advanced Technologies

Quality Control Inspector / Robotic Welding Technician Development

- Performed manufacturing quality inspections and supported fabrication quality requirements.
- Developed knowledge of welding quality, manufacturing documentation, inspection requirements, and production processes.
- Trained on robotic welding systems and automated welding processes.
- Worked directly with manufacturing, welding, quality, and engineering personnel to resolve production issues.
- Built the manufacturing and quality foundation later applied to Continuous Improvement engineering projects.

Manufacturing Operations

Production Coordinator / Team Leader

- Led an **assembly team of approximately 14 employees** and a **powder-coat team of approximately 5 employees**.
- Coordinated production requirements, staffing, workflow, priorities, and daily manufacturing activities.
- Led KPI discussions and worked to improve production performance.
- Participated in continuous improvement and Kaizen activities.
- Developed practical leadership experience managing personnel in a production environment.
- Supported problem solving involving quality, production flow, staffing, material, and scheduling.
- Worked closely with operators and management to maintain production requirements.

Manufacturing Operations

Assembly Floater / Manufacturing Operator / Backup Cell Leader

- Worked across multiple manufacturing operations and production areas.
- Developed experience with assembly processes, steel finishing, powder-coat finishing, quality requirements, and manufacturing workflow.
- Operated and supported automated production equipment including **pick-and-place robotics**
- Served as backup cell leader and assisted with production coordination and operator support.
- Developed hands-on understanding of manufacturing processes that now supports engineering and automation decisions.

ADDITIONAL EXPERIENCE

Graphic Design & Product Development

- Developed merchandise and product designs for artists.
- Coordinated production with overseas manufacturing suppliers.
- Worked with products including clothing, enamel/art pins, bags, and other custom merchandise.
- Gained experience communicating specifications, reviewing production requirements, and coordinating manufactured products.

Earlier Career Experience

Additional experience includes kitchen operations and hotel maintenance, contributing to a broad background in equipment, facilities, teamwork, customer service, and fast-paced operational environments.

EDUCATION

Associate Degree — Engineering Technology

Expected November 2027

Currently completing an Engineering Technology degree while working professionally in Continuous Improvement Engineering and manufacturing technology development.

Relevant areas of study include:

- Engineering Technology
 - Manufacturing
 - Information Technology
 - Computer Systems
 - Automation
 - Artificial Intelligence
 - Operating Systems
 - Technical Problem Solving
-

TECHNICAL SKILLS

Programming & Development

Python • REST APIs • SQL • SQLite • Web Applications • Windows Applications • Android Applications • API Integration • Database Design • Automation Scripts

AI & Machine Learning

Large Language Models • Vision-Language Models • Local AI Deployment • Ollama • Qwen • AI Agents • Retrieval-Based AI • Prompt Engineering • GPU Compute • AI Workflow Orchestration

Manufacturing Systems

Epicor ERP • MES Concepts • Digital Work Instructions • Routing Analysis • BOM Analysis • Labor Tracking • Material Tracking • Production Scheduling • Quality Systems

Industrial Technology

Raspberry Pi • ESP32 • Barcode Systems • Sensors • Industrial IoT • Robotics • Autonomous Mobile Robots • Machine Monitoring

Continuous Improvement

Six Sigma • Kaizen • KPI Analysis • Time Studies • Process Mapping • Waste Reduction • Root Cause Analysis • Standard Work • Workflow Optimization

Manufacturing

Sheet Metal Fabrication • Assembly • Welding • Robotic Welding • Powder Coating • Quality Inspection • Production Coordination • Material Handling

SELECTED PROFESSIONAL STRENGTHS

- Combines **hands-on manufacturing experience with software and engineering knowledge.**
 - Able to translate shop-floor problems into practical digital or automated solutions.
 - Comfortable working directly with operators as well as engineers, IT personnel, managers, and executives.
 - Experience independently conceptualizing, developing, testing, and implementing manufacturing applications.
 - Strong understanding of both the technical and human sides of manufacturing technology adoption.
 - Focused on using AI and automation to **support employees and improve processes rather than automate without operational understanding.**
 - Experienced in building technology specifically around existing manufacturing workflows instead of forcing manufacturing operations to adapt to generic software.
-

CAREER FOCUS

Continuous Improvement Engineering • Manufacturing Engineering • Digital Manufacturing • Industrial Automation • Manufacturing Software • AI Solutions • Smart Manufacturing • Industry 4.0